Analysis of the DBSCAN and K-DBSCAN Algorithms

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This report provides fully reproducible code for the DBSCAN and K-DBSCAN analysis. It includes the manual implementations of the algorithms, how the simulated datasets were created, and the comparisons for execution time and clustering quality (DB and SI metrics). The document includes all figures and tables used in the final report.

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1 DBSCAN Algorithm

This section defines the manual implementation of the DBSCAN algorithm. It utilizes a queue in data structure and a pairwise distance matrix to handle cluster expansion efficiently in R.

```
manual_dbscan <- function(data, eps, minPts) {
  n <- nrow(data)
  cluster_assignment <- rep(0, n)  # 0 = unvisited, -1 = noise
  cluster_id <- 0

  # Compute pairwise Euclidean distances to define epsilon-neighborhoods
  dist_mat <- as.matrix(dist(data, method = "euclidean"))

  for (i in 1:n) {
    # Skip point if it has already been visited
    if (cluster_assignment[i] != 0) next

    # Retrieve the epsilon-neighborhood for point i
    neighbors <- which(dist_mat[i, ] <= eps)

    # Check core point condition
    if (length(neighbors) < minPts) {
      cluster_assignment[i] <- -1 # Temporarily classify as a noise point
    } else {
      # Core point verified: create a new density-based cluster
      cluster_id <- cluster_id + 1
      cluster_assignment[i] <- cluster_id

      # Identify directly density-reachable points to start the chain
      seeds <- setdiff(neighbors, i)

      # Initialize a queue to establish density-connected points
      queue <- integer(n)
      queue_len <- length(seeds)

      if (queue_len > 0) {
        queue[1:queue_len] <- seeds
      }

      head <- 1
      tail <- queue_len
```

```r
      # Expand the cluster to satisfy the Maximality requirement
      while (head <= tail) {
        current_p <- queue[head]
        head <- head + 1

        # If previously classified as noise, it is a border point
        if (cluster_assignment[current_p] == -1) {
          cluster_assignment[current_p] <- cluster_id
        }

        # If unvisited, process for density-reachability
        if (cluster_assignment[current_p] == 0) {
          cluster_assignment[current_p] <- cluster_id

          # Retrieve the epsilon-neighborhood of the current point
          new_neighbors <- which(dist_mat[current_p, ] <= eps)

          # If this point is also a core point, expand the reachable chain
          if (length(new_neighbors) >= minPts) {
            # Extract only unvisited or noise points
            valid_new <- new_neighbors[cluster_assignment[new_neighbors] <= 0]
            num_new <- length(valid_new)

            if (num_new > 0) {
              queue[(tail + 1):(tail + num_new)] <- valid_new
              tail <- tail + num_new
            }
          }
        }
      }
    }
  }
  return(cluster_assignment)
}
```

2 Simulation of Datasets

The simulation evaluates the algorithm's performance on non-linear geometries (sine curves, rings, and arcs) and its ability to distinguish clusters from background noise (Kuchciak, 2024). We demonstrate the advantages of DBSCAN clustering in comparison to k-means clustering.

```r
library(dbscan)
```

```
Attaching package: 'dbscan'

The following object is masked from 'package:stats':

    as.dendrogram
```

```r
library(ggplot2)
library(MASS)

set.seed(2201)

# Define the cluster centers and sizes
centers <- matrix(c(25, 25, -25, -25, 25, -20, -20, 20, -3, -7), ncol = 2, byrow = TRUE)
size <- 1000

# Generate synthetic clusters with different characteristics
cluster1 <- mvrnorm(n = size, mu = centers[1,], Sigma = matrix(c(9, 2, 2, 9), ncol = 2))
cluster2 <- mvrnorm(n = size, mu = centers[2,], Sigma = matrix(c(12, 0.5, 0.5, 4), ncol = 2))
cluster3 <- mvrnorm(n = size, mu = centers[3,], Sigma = matrix(c(6, 0.3, 0.3, 18), ncol = 2))
cluster4 <- mvrnorm(n = size, mu = centers[4,], Sigma = matrix(c(15, 0.1, 0.1, 15), ncol = 2
cluster5 <- mvrnorm(n = size, mu = centers[5,], Sigma = matrix(c(7, 1.3, 1.3, 5), ncol = 2))

# Cluster 6: Sine curve distribution
x_values <- seq(-40, -20, length.out = 1000)
y_values <- sin(x_values) * 2 + rnorm(1000, mean = 0, sd = 0.2) # Adding some noise to the si
cluster6 <- cbind(x_values, y_values)

# Cluster 7: Points along an arc
theta <- runif(1000, min = pi, max = 2 * pi)
r <- 5
cluster7 <- cbind(r * cos(theta) + 5, r * sin(theta) + 15)

# Cluster 8: Points in a "moon" shape
theta <- runif(1000, min = 0, max = pi)
r <- 10 + runif(1000, min = -1, max = 1)
cluster8 <- cbind(r * cos(theta) + 30, r * sin(theta))

# Cluster 9: Points in a "ring" shape around all other clusters
```

```r
theta <- runif(1000, min = 0, max = 2 * pi)
r_inner <- 48
r_outer <- 52
r <- sqrt(runif(1000, min = r_inner^2, max = r_outer^2))
cluster9 <- cbind(r * cos(theta), r * sin(theta))

# Noise and Outliers
noise <- cbind(runif(250, -60, 60), runif(250, -60, 60))
outliers <- cbind(rnorm(100, 0, 20), rnorm(100, 0, 20))

data_full <- rbind(cluster1, cluster2, cluster3, cluster4, cluster5,
                   cluster6, cluster7, cluster8, cluster9, noise, outliers)
df_full <- as.data.frame(data_full)
colnames(df_full) <- c("X", "Y")
```

2.1 Execution and Comparison

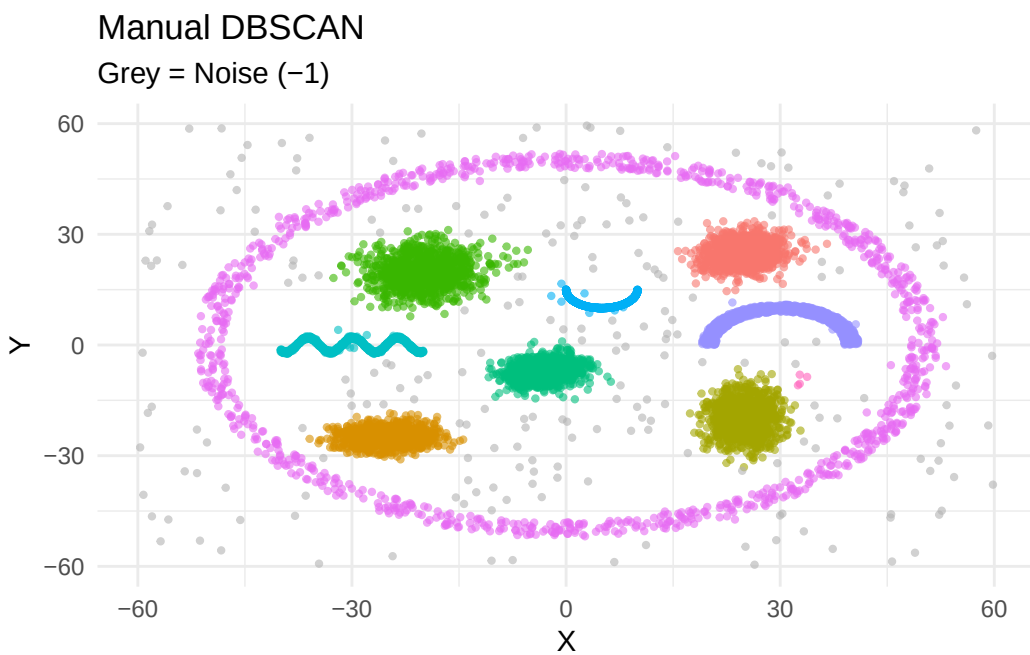
```r
set.seed(2201)

# Run Manual DBSCAN
results_dbscan <- manual_dbscan(df_full, eps = 2.5, minPts = 4)
df_full$Cluster <- as.factor(results_dbscan)
```

```r
# Run K-means for Comparison
km_result <- kmeans(df_full, centers = 10, nstart = 25)
df_full$Kmeans_Cluster <- as.factor(km_result$cluster)
```

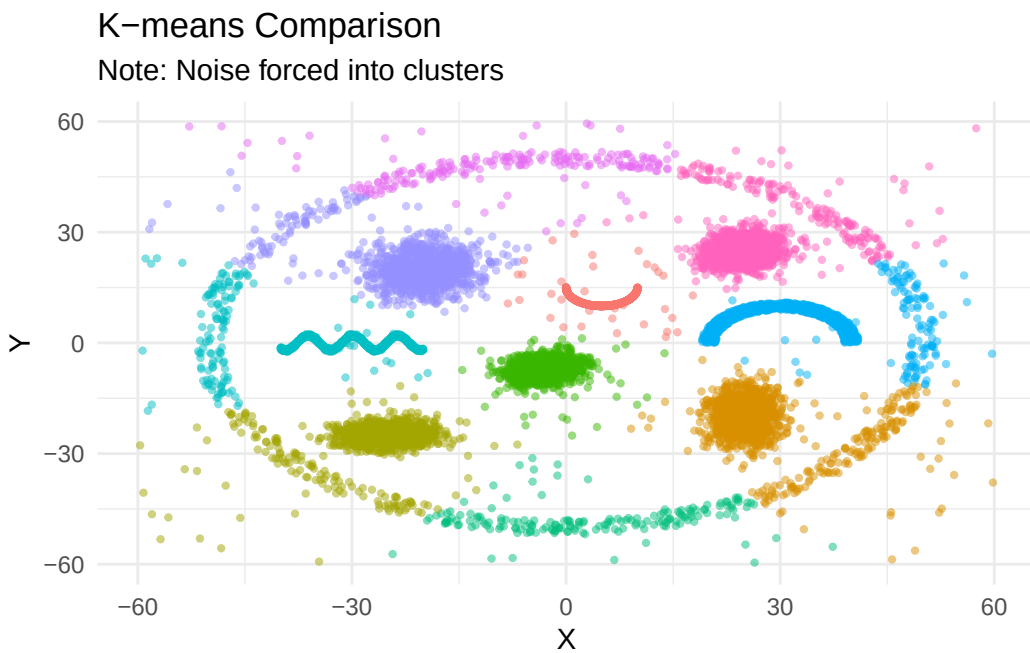
2.2 Visualization

```r
# Extract unique clusters excluding the noise (-1)
db_clusters <- unique(results_dbscan[results_dbscan != -1])
db_colors <- setNames(scales::hue_pal()(length(db_clusters)), db_clusters)
db_palette <- c("-1" = "grey70", db_colors)


# DBSCAN Results
ggplot(df_full, aes(x = X, y = Y, color = Cluster)) +
  geom_point(alpha = 0.6, size = 0.8) +
```

```r
  scale_color_manual(values = db_palette) +
  theme_minimal() +
  labs(title = "Manual DBSCAN", subtitle = "Grey = Noise (-1)") +
  theme(legend.position = "none")
```




```r
# K-means Results
ggplot(df_full, aes(x = X, y = Y, color = Kmeans_Cluster)) +
  geom_point(alpha = 0.5, size = 0.8) +
  theme_minimal() +
  labs(title = "K-means Comparison", subtitle = "Note: Noise forced into clusters") +
  theme(legend.position = "none")
```



The comparison above highlights the advantages of DBSCAN. It identified clusters of arbitrary shape and isolated noise points. In contrast, standard K-means forces noise into the nearest cluster, distorting the centroids and the overall structure of the data. K-means also notably did not pick up the shape of the elliptical ring around the other clusters.

3 K-means++ Algorithm

K-means++ improves upon the standard k-means algorithm by providing a specific method for choosing initial cluster centers. Standard k-means often relies on random initialization, which can lead to poor local optima. The k-means++ approach spreads out the initial centers before starting the iterative assignment and update steps.

3.1 Implementation

The following implementation includes the probabilistic seeding process and the iterative refinement loop.

```
# Function to compute Euclidean distance between a point and a set of centers
get_min_dist_sq <- function(data, centers) {
  # Calculate squared distances from each point to the nearest existing center
  apply(data, 1, function(point) {
    min(colSums((t(centers) - point)^2))
```

```r
  })
}

manual_kmeans_pp <- function(data, k, max_iter = 100) {
  data_mat <- as.matrix(data)
  n <- nrow(data_mat)

  # --- Step 1: K-means++ Seeding ---

  # Choose the first center uniformly at random
  centers <- matrix(NA, nrow = k, ncol = ncol(data_mat))
  centers[1, ] <- data_mat[sample(1:n, 1), ]

  if (k > 1) {
    for (i in 2:k) {
      # Compute squared distances to the nearest center already chosen
      dist_sq <- get_min_dist_sq(data_mat, centers[1:(i-1), , drop = FALSE])

      # Select next center with probability proportional to D(x)^2
      # This ensures centers are spread out across the feature space
      probs <- dist_sq / sum(dist_sq)
      next_idx <- sample(1:n, 1, prob = probs)
      centers[i, ] <- data_mat[next_idx, ]
    }
  }

  # --- Step 2: Standard K-means Iteration ---

  assignments <- rep(0, n)
  prev_assignments <- rep(-1, n)
  iter <- 0

  while (!all(assignments == prev_assignments) && iter < max_iter) {
    prev_assignments <- assignments
    iter <- iter + 1

    # Assignment Step: Assign each point to the nearest center
    dist_to_centers <- as.matrix(dist(rbind(centers, data_mat)))
    # Extract distances between points and centers only
    dist_matrix <- dist_to_centers[(k + 1):(n + k), 1:k]
    assignments <- apply(dist_matrix, 1, which.min)
```

```
    # Update Step: Recompute centers as the mean of assigned points
    for (j in 1:k) {
      if (sum(assignments == j) > 0) {
        centers[j, ] <- colMeans(data_mat[assignments == j, , drop = FALSE])
      }
    }
  }

  return(list(cluster = assignments, centers = centers, iterations = iter))
}
```

3.2 Simulation and Evaluation

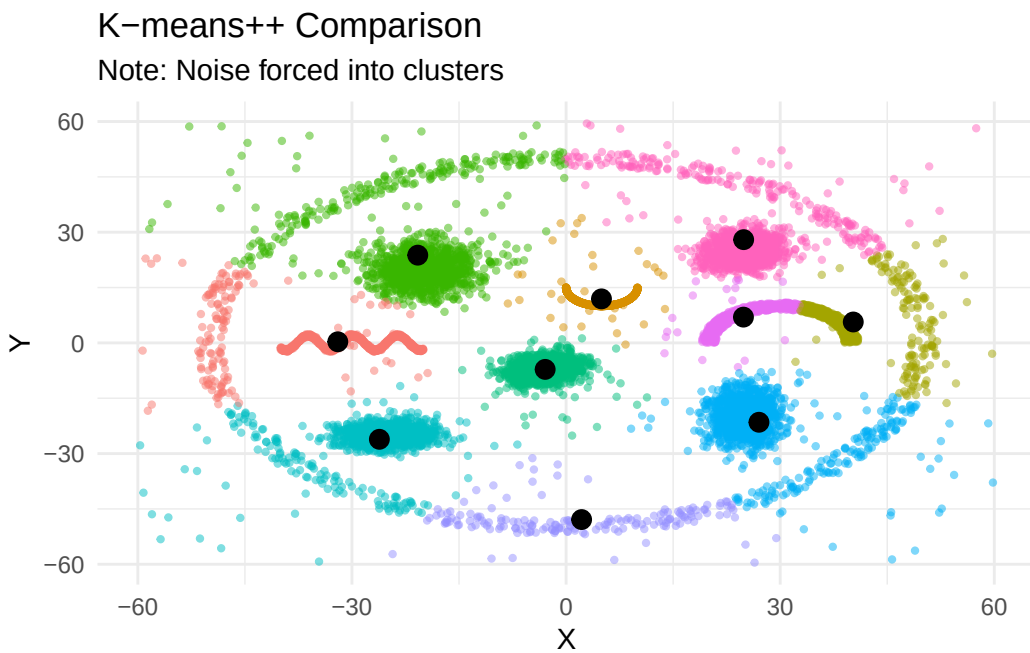
We reuse the previous synthetic dataset to evaluate how effectively k-means++ identifies the underlying structure.

```
set.seed(2201)
# Run Manual K-means++
# We choose k=10 to match the number of intended clusters in the generation
k_clusters <- 10
results_kmpp <- manual_kmeans_pp(df_full[, c("X", "Y")], k = k_clusters)
df_full$Kmpp_Cluster <- as.factor(results_kmpp$cluster)

centers_df <- as.data.frame(results_kmpp$centers)
colnames(centers_df) <- c("X", "Y")
```

3.3 Visualization

The plot below displays the final cluster assignments. The larger black points represent the final centroids calculated by the algorithm.

```
# Kmpp-means Results
ggplot(df_full, aes(x = X, y = Y, color = Kmpp_Cluster)) +
  geom_point(alpha = 0.5, size = 0.8) +
  geom_point(data = centers_df, aes(x = X, y = Y),
             color = "black", size = 3) +
  theme_minimal() +
  labs(title = "K-means++ Comparison", subtitle = "Note: Noise forced into clusters") +
  theme(legend.position = "none")
```



4 K-DBSCAN Algorithm

The primary goal of the K-DBSCAN hybrid approach is to process data in localized segments and then unify density-connected clusters that cross the boundaries of those segments. By partitioning the space with K-means, the algorithm only performs exhaustive distance checks between points within the same group or between adjacent groups.

4.1 Implementation

The following implementation integrates the `manual_dbscan` and `manual_kmeans_pp` methods.

```
single_link_dist <- function(clu_1, clu_2) {
  clu_1 <- as.matrix(clu_1)
  clu_2 <- as.matrix(clu_2)

  if (nrow(clu_1) == 0 || nrow(clu_2) == 0) {
    return(Inf)
  }

  dist_mat <- as.matrix(dist(rbind(clu_1, clu_2)))
```

```r
  n1 <- nrow(clu_1)
  n2 <- nrow(clu_2)
  return(min(dist_mat[1:n1, (n1 + 1):(n1 + n2), drop = FALSE]))
}

cluster_dist <- function(clu_1, clu_2) {
  clu_1 <- as.matrix(clu_1)
  clu_2 <- as.matrix(clu_2)

  if (nrow(clu_1) == 0 || nrow(clu_2) == 0) {
    return(Inf)
  }

  center_1 <- colMeans(clu_1)
  center_2 <- colMeans(clu_2)

  dis_KCij <- sqrt(sum((center_1 - center_2)^2))
  dis_ri <- max(sqrt(rowSums((clu_1 - matrix(center_1, nrow = nrow(clu_1), ncol = ncol(clu_1
  dis_rj <- max(sqrt(rowSums((clu_2 - matrix(center_2, nrow = nrow(clu_2), ncol = ncol(clu_2

  return(dis_KCij - dis_ri - dis_rj)
}

k_dbscan <- function(D, k, max_iter, eps, minPts) {
  # Step 1: Normalization of D
  D_mat <- as.matrix(scale(D))
  n <- nrow(D_mat)

  if (n == 0) {
    return(integer(0))
  }

  # Step 2: Data grouping with K-means Algorithm
  km_results <- manual_kmeans_pp(D_mat, k = k, max_iter = max_iter)
  group_id <- km_results$cluster

  # Step 3: Run DBSCAN inside each K-means group
  group_points <- vector("list", k)
  numclu <- integer(k)

  for (i in seq_len(k)) {
    group_idx <- which(group_id == i)
```

```r
  if (length(group_idx) == 0) {
    group_points[[i]] <- list()
    next
  }

  group_data <- D_mat[group_idx, , drop = FALSE]
  local_labels <- manual_dbscan(group_data, eps, minPts)
  local_cluster_ids <- sort(unique(local_labels[local_labels > 0]))

  numclu[i] <- length(local_cluster_ids)
  group_points[[i]] <- lapply(local_cluster_ids, function(local_cluster) {
    group_idx[local_labels == local_cluster]
  })
}

# Step 4: Merge border clusters
flag <- vector("list", k)
for (i in seq_len(k)) {
  flag[[i]] <- rep(0L, numclu[i])
}

counter <- 1L

for (a in seq_len(k - 1L)) {
  for (b in seq_len(k)) {
    if (numclu[a] == 0L || numclu[b] == 0L) next

    dis_k <- cluster_dist(D_mat[group_id == a, , drop = FALSE],
                          D_mat[group_id == b, , drop = FALSE])
    if (dis_k > eps) next

    for (c in seq_len(numclu[a])) {
      for (d in seq_len(numclu[b])) {
        dis_db <- single_link_dist(
          D_mat[group_points[[a]][[c]], , drop = FALSE],
          D_mat[group_points[[b]][[d]], , drop = FALSE]
        )

        if (dis_db <= eps) {
          if (flag[[a]][c] == 0L && flag[[b]][d] == 0L) {
            flag[[a]][c] <- counter
            flag[[b]][d] <- counter
```

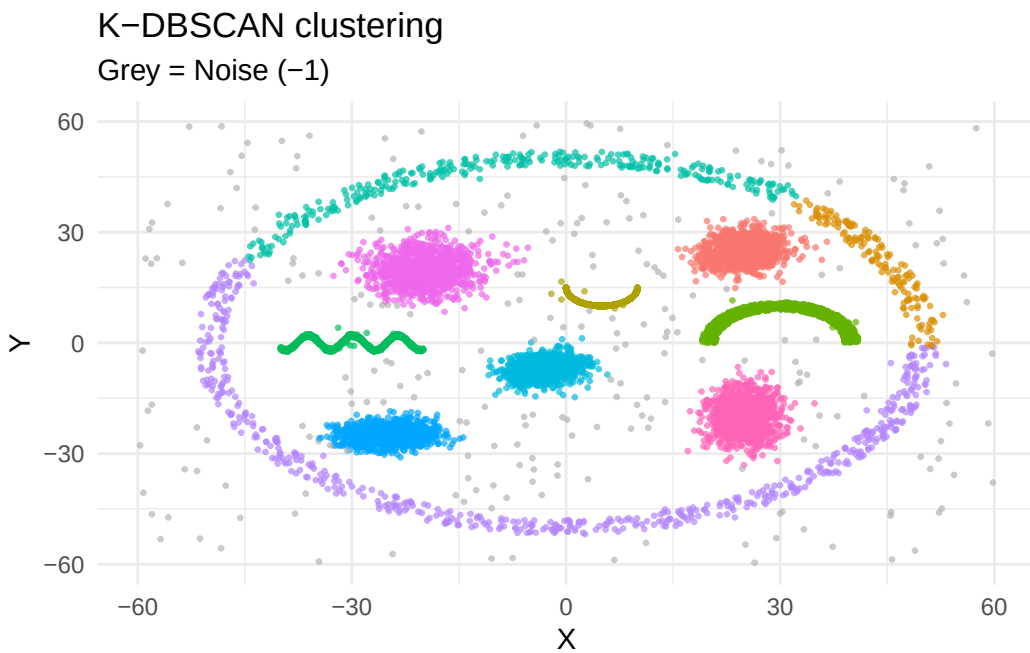
```
            counter <- counter + 1L
          } else {
            new_flag <- max(flag[[a]][c], flag[[b]][d])
            flag[[a]][c] <- new_flag
            flag[[b]][d] <- new_flag
          }
        }
      }
    }
  }
}

for (i in seq_len(k)) {
  if (numclu[i] == 0L) next
  for (c in seq_len(numclu[i])) {
    if (flag[[i]][c] == 0L) {
      flag[[i]][c] <- counter
      counter <- counter + 1L
    }
  }
}

# Step 5: Run DBSCAN again inside each flagged component
final_labels <- rep(-1L, n)
final_cluster_id <- 0L

for (i in seq_len(counter - 1L)) {
  component_points <- integer(0)

  for (g in seq_len(k)) {
    if (numclu[g] == 0L) next
    for (c in seq_len(numclu[g])) {
      if (flag[[g]][c] == i) {
        component_points <- c(component_points, group_points[[g]][[c]])
      }
    }
  }

  component_points <- sort(unique(component_points))
  if (length(component_points) == 0) next

  flagged_data <- D_mat[component_points, , drop = FALSE]
```

```r
    flagged_result <- manual_dbscan(flagged_data, eps, minPts)

    for (local_cluster in sort(unique(flagged_result[flagged_result > 0]))) {
      final_cluster_id <- final_cluster_id + 1L
      final_labels[component_points[flagged_result == local_cluster]] <- final_cluster_id
    }
  }

  return(final_labels)
}
```

4.2 Application on Simulated Data

This section applies the K-DBSCAN algorithm to the synthetic data from earlier.

```r
results_kdbscan <- k_dbscan(df_full[, c("X", "Y")], 10, 4, 0.12, 5)
df_full$kdbs_cluster <- factor(results_kdbscan)

real_clusters <- sort(unique(results_kdbscan[results_kdbscan > 0]))

ggplot(df_full, aes(x = X, y = Y, color = kdbs_cluster)) +
  geom_point(size = 0.5, alpha = 0.7) +
  scale_color_manual(
    values = c("-1" = "grey70",
               setNames(scales::hue_pal()(length(real_clusters)),
                        as.character(real_clusters)))
  ) +
  labs(title = "K-DBSCAN clustering", subtitle = "Grey = Noise (-1)") +
  theme_minimal() +
  theme(legend.position = "none")
```



5 Comparison of DBSCAN and K-DBSCAN

The authors noted the challenge of running these algorithms on large datasets, citing computational capacity as a limit. For our cases, to show if there's any improvement in DBSCAN compared to the K-DBSCAN algorithm, we rewrite our above implementations with some changes. To enable our computers to run the algorithms on larger datasets, we optimize some of the key calculations that were originally so intensive, that it was beyond our computer's computing power. In specific, the distance matrix calculations for each point gets unmanageable once we reach a dataset of higher than 10,000 points.

Below are some re-wired versions of the coded algorithms. We use this to enable our computer to run the techniques, while still keeping the formulation of the algorithm the same.

5.1 Optimized Algorithms for Computation

```
find_uf <- function(parent, x) {
  while (parent[x] != x) x <- parent[x]
  return(x)
}
# swapped out your old flag/counter system in step 4 for a union-find approach.
# makes merging the border clusters way cleaner without needing massive loops.
```

```r
union_uf <- function(parent, x, y) {
  rx <- find_uf(parent, x)
  ry <- find_uf(parent, y)
  if (rx != ry) parent[rx] <- ry
  return(parent)
}

# added this to find the max reach (radius) of a cluster.
# we use this later to skip distance checks for clusters that are physically far
get_radius <- function(data_mat, center_vec) {
  if (nrow(data_mat) == 0) return(0)
  diffs <- sweep(data_mat, 2, center_vec, "-")
  return(max(sqrt(rowSums(diffs^2))))
}

# this completely replaces single_link_dist and as.matrix(dist()).
# uses a linear algebra trick: (A-B)^2 = A^2 + B^2 - 2AB to compute distances
# across matrices  without creating a giant N x N object that kills the RAM.
fast_min_dist <- function(A, B) {
  A <- as.matrix(A)
  B <- as.matrix(B)
  if (nrow(A) == 0 || nrow(B) == 0) return(Inf)

  D2 <- outer(rowSums(A^2), rowSums(B^2), "+") - 2 * tcrossprod(A, B)
  D2[D2 < 0] <- 0
  return(sqrt(min(D2)))
}

manual_dbscan_opt <- function(D, eps, minPts) {
  D <- as.matrix(D)
  n <- nrow(D)
  if (n == 0) return(integer(0))

  labels <- rep(0L, n)
  cluster_id <- 0L

  # rather than calling dist() on everything upfront, we calculate distances
  # point-by-point. this is what actually prevents the 16GB memory crash
  # during the dbscan phase.
  for (i in seq_len(n)) {
    if (labels[i] != 0L) next
```

```r
    diffs <- sweep(D, 2, D[i, ], "-")
    dists <- sqrt(rowSums(diffs^2))
    neighbors <- which(dists <= eps)

    if (length(neighbors) < minPts) {
      labels[i] <- -1L
    } else {
      cluster_id <- cluster_id + 1L
      labels[i] <- cluster_id
      seeds <- setdiff(neighbors, i)

      while (length(seeds) > 0) {
        cp <- seeds[1]
        seeds <- seeds[-1]

        if (labels[cp] == -1L) labels[cp] <- cluster_id
        if (labels[cp] != 0L) next

        labels[cp] <- cluster_id
        diffs_seed <- sweep(D, 2, D[cp, ], "-")
        dists_seed <- sqrt(rowSums(diffs_seed^2))
        seed_neighbors <- which(dists_seed <= eps)

        if (length(seed_neighbors) >= minPts) {
          new_seeds <- setdiff(seed_neighbors, c(seeds, i, cp))
          if (length(new_seeds) > 0) seeds <- c(seeds, new_seeds)
        }
      }
    }
  }
  return(labels)
}

k_dbscan_opt <- function(D, eps, minPts) {
  D_mat <- as.matrix(scale(D))
  n <- nrow(D_mat)
  if (n < 2) return(rep(-1L, n))

  k <- max(1, min(floor(sqrt(n)), n - 1))

  # swapped your manual_kmeans_pp for the base stats package kmeans.
  # running manual k-means in an interpreted loop was just adding unnecessary
```

```r
# time overhead.
km <- suppressWarnings(kmeans(D_mat, centers = k, iter.max = 4))
group_id <- km$cluster
centers <- km$centers

group_points <- vector("list", k)
numclu <- integer(k)
group_radii <- numeric(k)
sub_centers <- vector("list", k)
sub_radii <- vector("list", k)

for (i in seq_len(k)) {
  idx <- which(group_id == i)
  if (length(idx) == 0) next

  group_data <- D_mat[idx, , drop = FALSE]

  # calculating the boundary radius for the whole k-means partition here to
  # prune checks later
  group_radii[i] <- get_radius(group_data, centers[i, ])

  local_labels <- manual_dbscan_opt(group_data, eps, minPts)
  u_labels <- sort(unique(local_labels[local_labels > 0]))
  numclu[i] <- length(u_labels)

  group_points[[i]] <- lapply(u_labels, function(l) idx[local_labels == l])
  sub_centers[[i]] <- lapply(group_points[[i]], function(p) colMeans(D_mat[p, , drop = FALS

  # storing boundary radii for the internal sub-clusters too
  sub_radii[[i]] <- mapply(function(p, ctr) get_radius(D_mat[p, , drop = FALSE], ctr),
                           group_points[[i]], sub_centers[[i]])
}

total_subclu <- sum(numclu)
if (total_subclu == 0) return(rep(-1L, n))

parent <- seq_len(total_subclu)

sub_id_map <- vector("list", k)
curr <- 0
for (g in seq_len(k)) {
  if(numclu[g] > 0) {
```

```
    sub_id_map[[g]] <- (curr + 1):(curr + numclu[g])
    curr <- curr + numclu[g]
  }
}

# optimization: instead of checking every group against every other group,
# we only check groups whose maximum radii actually overlap.
g_dist <- as.matrix(dist(centers))
g_adj <- (g_dist - outer(group_radii, group_radii, "+")) <= eps
g_adj[lower.tri(g_adj, diag = TRUE)] <- FALSE
pairs <- which(g_adj, arr.ind = TRUE)

if (nrow(pairs) > 0) {
  for (idx in seq_len(nrow(pairs))) {
    g1 <- pairs[idx, 1]
    g2 <- pairs[idx, 2]

    for (c1 in seq_len(numclu[g1])) {
      for (c2 in seq_len(numclu[g2])) {
        d_dc <- sqrt(sum((sub_centers[[g1]][[c1]] - sub_centers[[g2]][[c2]])^2))

        # secondary prune: even if the k-means groups overlap, we only run the
        # expensive distance calculation if the specific sub-clusters inside them
        # overlap too.
        if ((d_dc - (sub_radii[[g1]][c1] + sub_radii[[g2]][c2])) <= eps) {

          # using the fast matrix math here instead of single_link_dist
          if (fast_min_dist(D_mat[group_points[[g1]][[c1]], ], D_mat[group_points[[g2]][[c2
            parent <- union_uf(parent, sub_id_map[[g1]][c1], sub_id_map[[g2]][c2])
          }

        }
      }
    }
  }
}

# resolving final labels using the union-find structure instead of running a
# completely new pass of manual_dbscan on flagged components.
roots <- unique(vapply(seq_len(total_subclu), find_uf, integer(1), parent = parent))
final_labels <- rep(-1L, n)
cid <- 0
```

```r
  for (r in roots) {
    pts <- unlist(lapply(seq_len(k), function(g) {
      if(numclu[g] > 0) {
        matches <- which(vapply(sub_id_map[[g]], find_uf, 1, parent=parent) == r)
        if (length(matches) > 0) return(unlist(group_points[[g]][matches]))
      }
      return(NULL)
    }))

    if (length(pts) > 0) {
      cid <- cid + 1
      final_labels[pts] <- cid
    }
  }
  return(final_labels)
}
```

```r
find_uf <- function(parent, x) {
  while (parent[x] != x) x <- parent[x]
  return(x)
}

union_uf <- function(parent, x, y) {
  rx <- find_uf(parent, x)
  ry <- find_uf(parent, y)
  if (rx != ry) parent[rx] <- ry
  return(parent)
}

# Safely calculate the max distance from a center (cluster radius)
get_radius <- function(data_mat, center_vec) {
  if (nrow(data_mat) == 0) return(0)
  diffs <- sweep(data_mat, 2, center_vec, "-")
  return(max(sqrt(rowSums(diffs^2))))
}

# Fast cross-distance using linear algebra to avoid R loops
fast_min_dist <- function(A, B) {
  A <- as.matrix(A)
  B <- as.matrix(B)
  if (nrow(A) == 0 || nrow(B) == 0) return(Inf)
```

```r
  D2 <- outer(rowSums(A^2), rowSums(B^2), "+") - 2 * tcrossprod(A, B)
  D2[D2 < 0] <- 0
  return(sqrt(min(D2)))
}

# --- 2. Memory-Safe Manual DBSCAN (The O(N^2) Baseline) ---
manual_dbscan_opt <- function(D, eps, minPts) {
  D <- as.matrix(D)
  n <- nrow(D)
  if (n == 0) return(integer(0))

  labels <- rep(0L, n)
  cluster_id <- 0L

  for (i in seq_len(n)) {
    if (labels[i] != 0L) next

    diffs <- sweep(D, 2, D[i, ], "-")
    dists <- sqrt(rowSums(diffs^2))
    neighbors <- which(dists <= eps)

    if (length(neighbors) < minPts) {
      labels[i] <- -1L
    } else {
      cluster_id <- cluster_id + 1L
      labels[i] <- cluster_id
      seeds <- setdiff(neighbors, i)

      while (length(seeds) > 0) {
        cp <- seeds[1]
        seeds <- seeds[-1]

        if (labels[cp] == -1L) labels[cp] <- cluster_id
        if (labels[cp] != 0L) next

        labels[cp] <- cluster_id
        diffs_seed <- sweep(D, 2, D[cp, ], "-")
        dists_seed <- sqrt(rowSums(diffs_seed^2))
        seed_neighbors <- which(dists_seed <= eps)

        if (length(seed_neighbors) >= minPts) {
          new_seeds <- setdiff(seed_neighbors, c(seeds, i, cp))
```

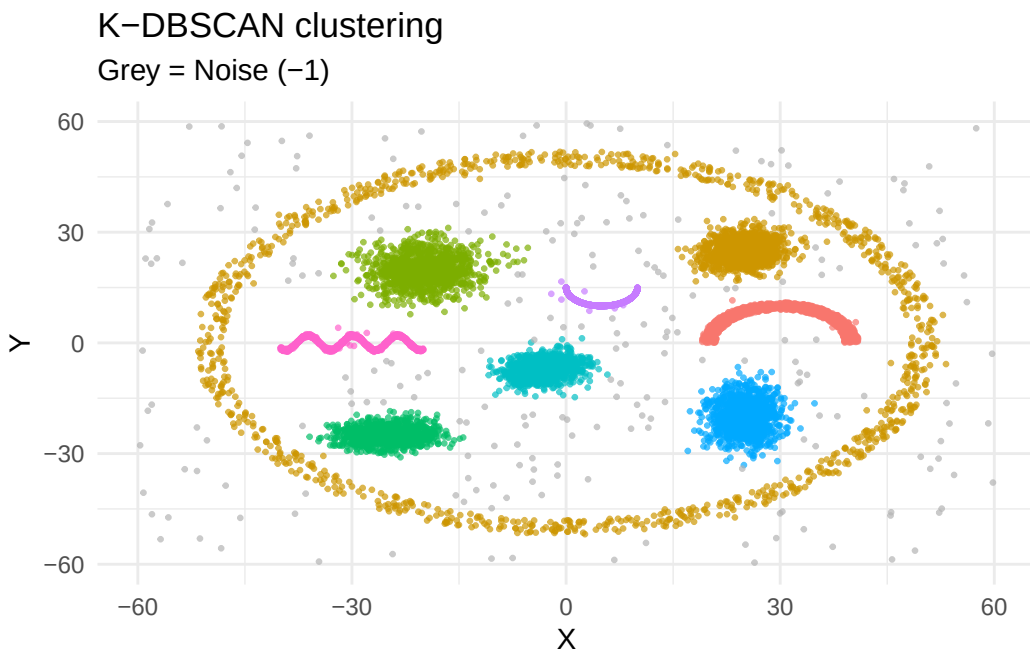
```
          if (length(new_seeds) > 0) seeds <- c(seeds, new_seeds)
        }
      }
    }
  }
  return(labels)
}

# --- 3. Optimized K-DBSCAN ---
k_dbscan_opt <- function(D, eps, minPts) {
  D_mat <- as.matrix(scale(D))
  n <- nrow(D_mat)
  if (n < 2) return(rep(-1L, n))

  k <- max(1, min(floor(sqrt(n)), n - 1))

  km <- suppressWarnings(kmeans(D_mat, centers = k, iter.max = 4))
  group_id <- km$cluster
  centers <- km$centers

  group_points <- vector("list", k)
  numclu <- integer(k)
  group_radii <- numeric(k)
  sub_centers <- vector("list", k)
  sub_radii <- vector("list", k)

  for (i in seq_len(k)) {
    idx <- which(group_id == i)
    if (length(idx) == 0) next

    group_data <- D_mat[idx, , drop = FALSE]
    group_radii[i] <- get_radius(group_data, centers[i, ])

    local_labels <- manual_dbscan_opt(group_data, eps, minPts)
    u_labels <- sort(unique(local_labels[local_labels > 0]))
    numclu[i] <- length(u_labels)

    group_points[[i]] <- lapply(u_labels, function(l) idx[local_labels == l])
    sub_centers[[i]] <- lapply(group_points[[i]], function(p) colMeans(D_mat[p, , drop = FALS
    sub_radii[[i]] <- mapply(function(p, ctr) get_radius(D_mat[p, , drop = FALSE], ctr),
                             group_points[[i]], sub_centers[[i]])
  }
```

```r
total_subclu <- sum(numclu)
if (total_subclu == 0) return(rep(-1L, n))
parent <- seq_len(total_subclu)

sub_id_map <- vector("list", k)
curr <- 0
for (g in seq_len(k)) {
  if(numclu[g] > 0) {
    sub_id_map[[g]] <- (curr + 1):(curr + numclu[g])
    curr <- curr + numclu[g]
  }
}

g_dist <- as.matrix(dist(centers))
g_adj <- (g_dist - outer(group_radii, group_radii, "+")) <= eps
g_adj[lower.tri(g_adj, diag = TRUE)] <- FALSE
pairs <- which(g_adj, arr.ind = TRUE)

if (nrow(pairs) > 0) {
  for (idx in seq_len(nrow(pairs))) {
    g1 <- pairs[idx, 1]
    g2 <- pairs[idx, 2]

    for (c1 in seq_len(numclu[g1])) {
      for (c2 in seq_len(numclu[g2])) {
        d_dc <- sqrt(sum((sub_centers[[g1]][[c1]] - sub_centers[[g2]][[c2]])^2))
        if ((d_dc - (sub_radii[[g1]][c1] + sub_radii[[g2]][c2])) <= eps) {

          if (fast_min_dist(D_mat[group_points[[g1]][[c1]], ], D_mat[group_points[[g2]][[c2
            parent <- union_uf(parent, sub_id_map[[g1]][c1], sub_id_map[[g2]][c2])
          }

        }
      }
    }
  }
}

roots <- unique(vapply(seq_len(total_subclu), find_uf, integer(1), parent = parent))
final_labels <- rep(-1L, n)
cid <- 0
```

```r
  for (r in roots) {
    pts <- unlist(lapply(seq_len(k), function(g) {
      if(numclu[g] > 0) {
        matches <- which(vapply(sub_id_map[[g]], find_uf, 1, parent=parent) == r)
        if (length(matches) > 0) return(unlist(group_points[[g]][matches]))
      }
      return(NULL)
    }))

    if (length(pts) > 0) {
      cid <- cid + 1
      final_labels[pts] <- cid
    }
  }
  return(final_labels)
}
```

The optimized 'k_dbscan' 's results do not vastly differ from the original implementation from above.

```r
results_kdbscan <- k_dbscan_opt(df_full[, c("X", "Y")], 0.12, 5)
df_full$kdbs_cluster <- factor(results_kdbscan)

real_clusters <- sort(unique(results_kdbscan[results_kdbscan > 0]))

ggplot(df_full, aes(x = X, y = Y, color = kdbs_cluster)) +
  geom_point(size = 0.5, alpha = 0.7) +
  scale_color_manual(
    values = c("-1" = "grey70",
               setNames(scales::hue_pal()(length(real_clusters)),
                        as.character(real_clusters)))
  ) +
  labs(title = "K-DBSCAN clustering", subtitle = "Grey = Noise (-1)") +
  theme_minimal() +
  theme(legend.position = "none")
```



5.2 Clustering Quality Metrics

```
library(cluster)
```

```
Warning: package 'cluster' was built under R version 4.5.2
```

```
library(clusterSim)

evaluate_packaged_metrics <- function(data, labels, max_sample = 5000) {
  # 1. Filter out noise points (-1) and unassigned points (0)
  valid_idx <- labels > 0

  if (sum(valid_idx) == 0 || length(unique(labels[valid_idx])) < 2) {
    return(c(DB = NA, SI = NA))
  }

  data_c <- data[valid_idx, , drop = FALSE]
  labels_c <- labels[valid_idx]

  # 2. Davies-Bouldin Index (FIXED: parameter is 'centrotypes')
  db_res <- index.DB(data_c, as.integer(labels_c), centrotypes = "centroids")
```

```r
  db_val <- db_res$DB

  # 3. Silhouette Index (Sampled for memory safety)
  if (nrow(data_c) > max_sample) {
    set.seed(42)
    sample_idx <- sample(seq_len(nrow(data_c)), max_sample)
    data_c <- data_c[sample_idx, ]
    labels_c <- labels_c[sample_idx]
  }

  dist_mat <- dist(data_c)
  sil_res <- silhouette(as.integer(labels_c), dist_mat)
  si_val <- mean(sil_res[, "sil_width"])

  return(c(DB = db_val, SI = si_val))
}
```

5.3 Executive Time and Quality Metrics

```r
# Simulation Parameters
test_sizes <- c(4000, 8000, 16000, 32000, 64000)
noise_ratio <- 0.10
eps_sim <- 0.15
minPts_sim <- 10 # Slightly higher to handle the 10% noise better

# Helper for constant noise density
get_noise_limit <- function(n) { 4 * sqrt(n / 4000) }


results_master <- data.frame()

# Integrated Loop
for (n in test_sizes) {
  cat("Processing N =", n, "---")

  # Generate Data
  set.seed(n)
  limit <- get_noise_limit(n)
  n_noise <- floor(n * noise_ratio)
  n_signal <- n - n_noise
```

```r
  pts_per_blob <- floor(n_signal / 4)

  blob_centers <- matrix(c(-2,-2, 2,2, -2,2, 2,-2), ncol = 2, byrow = TRUE)
  signal_data <- do.call(rbind, lapply(1:4, function(i) {
    matrix(rnorm(pts_per_blob * 2, mean = blob_centers[i,], sd = 0.4), ncol = 2)
  }))
  noise_data <- matrix(runif(n_noise * 2, min = -limit, max = limit), ncol = 2)

  sim_data <- scale(rbind(signal_data, noise_data))
  sim_data <- sim_data[sample(nrow(sim_data)), ] # Shuffle

  # Benchmark & Run Standard DBSCAN
  cat("\n  Running Standard DBSCAN... ")
  t_db <- system.time({
    res_db <- manual_dbscan_opt(sim_data, eps_sim, minPts_sim)
  })
  m_db <- evaluate_packaged_metrics(sim_data, res_db)

  # Benchmark & Run K-DBSCAN
  cat("Running K-DBSCAN... ")
  t_kdb <- system.time({
    res_kdb <- k_dbscan_opt(sim_data, eps_sim, minPts_sim)
  })
  m_kdb <- evaluate_packaged_metrics(sim_data, res_kdb)

  row_data <- data.frame(
    N = n,
    DBSCAN_Time = t_db["elapsed"],
    K_DBSCAN_Time = t_kdb["elapsed"],
    DBSCAN_DB = m_db["DB"],
    K_DBSCAN_DB = m_kdb["DB"],
    DBSCAN_SI = m_db["SI"],
    K_DBSCAN_SI = m_kdb["SI"]
  )
  results_master <- rbind(results_master, row_data)
}
```

```
Processing N = 4000 ---
  Running Standard DBSCAN... Running K-DBSCAN... Processing N = 8000 ---
  Running Standard DBSCAN... Running K-DBSCAN... Processing N = 16000 ---
  Running Standard DBSCAN... Running K-DBSCAN... Processing N = 32000 ---
  Running Standard DBSCAN... Running K-DBSCAN... Processing N = 64000 ---
```

```
Running Standard DBSCAN... Running K-DBSCAN...
```

```r
cat("EXECUTION PERFORMANCE & CLUSTER QUALITY REPORT\n")
```

```
EXECUTION PERFORMANCE & CLUSTER QUALITY REPORT
```

```r
print(results_master, row.names = FALSE)
```

```
     N DBSCAN_Time K_DBSCAN_Time DBSCAN_DB K_DBSCAN_DB DBSCAN_SI K_DBSCAN_SI
  4000       0.433         0.149 0.2025184   0.2015455 0.8730855   0.8735681
  8000       1.641         0.360 0.2048695   0.2028794 0.8717193   0.8732632
 16000       7.042         0.914 0.2025086   0.2016875 0.8732756   0.8737005
 32000      34.554         2.394 0.2919258   0.2167475 0.7272872   0.8075189
 64000     191.643         8.807 0.5985437   0.5713018 0.7318217   0.7618845
```

5.4 Visualizations

```r
df_lines <- data.frame(
  N = rep(results_master$N, 2),
  Time = c(results_master$DBSCAN_Time, results_master$K_DBSCAN_Time),
  Algorithm = factor(
    rep(c("Standard DBSCAN", "K-DBSCAN"), each = nrow(results_master)),
    levels = c("Standard DBSCAN", "K-DBSCAN")
  )
)

ggplot(df_lines, aes(x = N, y = Time, color = Algorithm, shape = Algorithm)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Standard DBSCAN" = "firebrick", "K-DBSCAN" = "dodgerblue4")
  scale_shape_manual(values = c("Standard DBSCAN" = 16, "K-DBSCAN" = 17)) +
  labs(
    title = "Comparing Standard DBSCAN vs K-DBSCAN for Large Datasets",
    x = "Number of Data Points (N)",
    y = "Execution Time (Seconds)"
  ) +
  scale_x_continuous(limits = c(min(df_lines$N), max(df_lines$N))) +
  scale_y_continuous(limits = c(0, max(df_lines$Time) * 1.1)) +
  theme_minimal() +
```

```
  theme(
    plot.title = element_text(face = "bold", size = 14),
    legend.position = c(0.15, 0.85), # Positions legend in the top-left
    legend.title = element_blank(),
    legend.background = element_rect(fill = "white", color = NA),
    panel.grid.major = element_line(color = "lightgray", linetype = "dotted")
  )
```

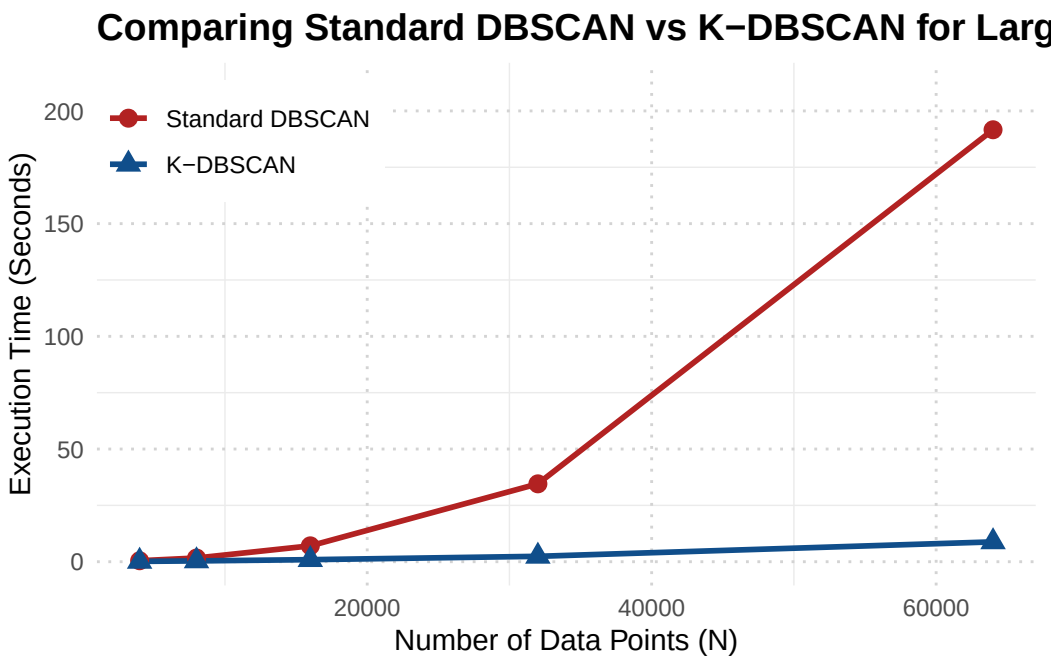
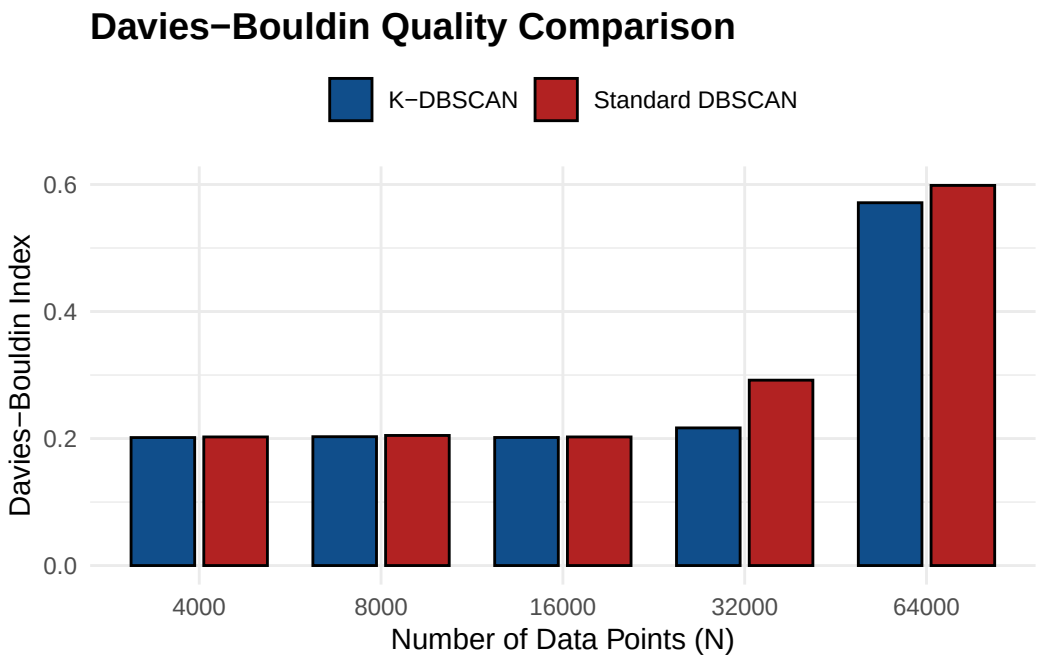


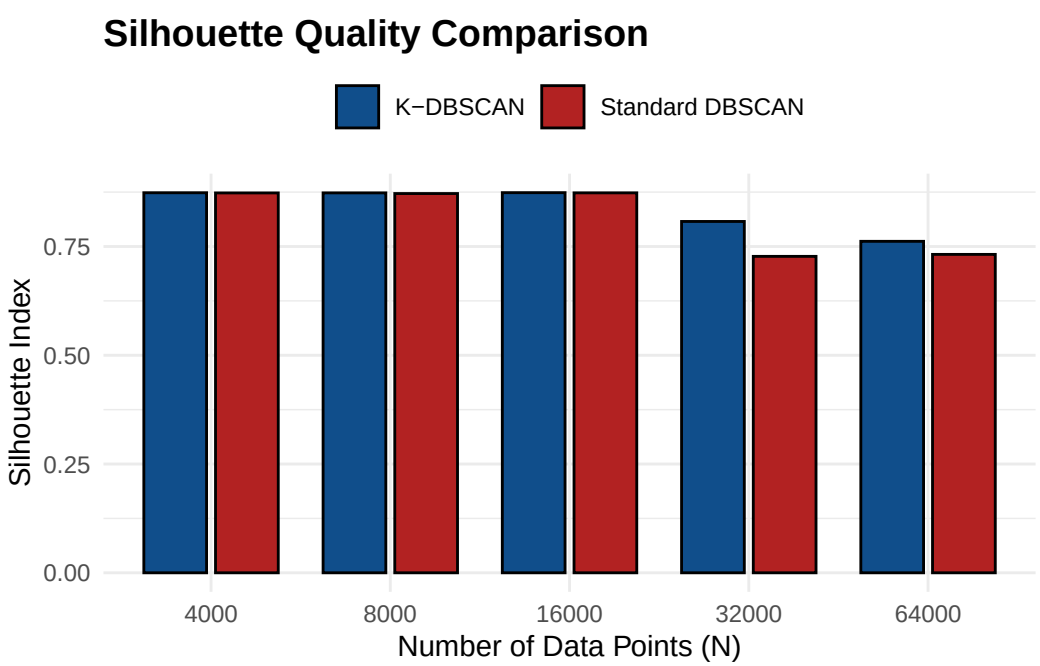
Figure 1: Diagram for comparison of execution time in the proposed method, K-DBSCAN, and standard DBSCAN.

```
# Davies-Bouldin Plot
df_db <- data.frame(
  N = rep(results_master$N, 2),
  Algorithm = rep(c("Standard DBSCAN", "K-DBSCAN"), each = nrow(results_master)),
  Score = c(results_master$DBSCAN_DB, results_master$K_DBSCAN_DB)
)

plot_db <- ggplot(df_db, aes(x = factor(N), y = Score, fill = Algorithm)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.8), width = 0.7, color="bla
  scale_fill_manual(values = c("Standard DBSCAN" = "firebrick", "K-DBSCAN" = "dodgerblue4"))
  labs(title = "Davies-Bouldin Quality Comparison",
       x = "Number of Data Points (N)",
```

```
       y = "Davies-Bouldin Index") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top",
        legend.title = element_blank())

# Silhouette Plot
df_si <- data.frame(
  N = rep(results_master$N, 2),
  Algorithm = rep(c("Standard DBSCAN", "K-DBSCAN"), each = nrow(results_master)),
  Score = c(results_master$DBSCAN_SI, results_master$K_DBSCAN_SI)
)

# Plot Silhouette
plot_si <- ggplot(df_si, aes(x = factor(N), y = Score, fill = Algorithm)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.8), width = 0.7, color="bla
  scale_fill_manual(values = c("Standard DBSCAN" = "firebrick", "K-DBSCAN" = "dodgerblue4"))
  labs(title = "Silhouette Quality Comparison",
       x = "Number of Data Points (N)",
       y = "Silhouette Index") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top",
        legend.title = element_blank())

plot_db
```


Davies−Bouldin Quality Comparison
K−DBSCAN
Standard DBSCAN
Davies−Bouldin Index
0.6
0.4
0.2
0.0
4000
8000
16000
32000
64000
Number of Data Points (N)

```
plot_si
```


Silhouette Quality Comparison
K−DBSCAN
Standard DBSCAN
Silhouette Index
0.75
0.50
0.25
0.00
4000
8000
16000
32000
64000
Number of Data Points (N)